

OBED MAVUNGU

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Education

Virginia Polytechnic Institute and State University

Bachelor of Engineering in Computer Science — GPA (4.0/4.0)

Expected May 2028

Blacksburg, Virginia

Relevant Coursework: Data Structures & Algorithms, Computer Organization, Discrete Math, Calculus III

Leadership: National Society of Black Engineers (NSBE), Church Youth Leader

Extracurricular: Amateur Boxing, Film Writer, Video Games

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, Java, C++, Swift, SQL

Frameworks & Libraries: React, Node.js, Express, Vite, Tailwind CSS, FastAPI, scikit-learn, SHAP, pandas, NumPy

Databases & ORM: PostgreSQL, Supabase, Prisma, Neon, PL/pgSQL, Row-Level Security

Tools & Platforms: Git, GitHub, Xcode, Vercel, Render, Capacitor, Postman, Jupyter

Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, REST APIs, Real-Time (WebSockets, Server-Sent Events), JWT Authentication, LLM Integration & Prompt Engineering

Work Experience

Zarnite (AI Startup)

June 2026 – Present

Full Stack Developer Intern — Engineering / AI Systems

Remote

- Build and maintain scalable full-stack web applications across a React/Next.js front-end and a Node.js, Express, and Python back-end, integrating REST APIs, databases (PostgreSQL, MongoDB), and cloud services.
- Contribute to AI/ML development by assisting with fine-tuning LLMs and classification models and deploying them to production using embeddings, vector search, and inference serving.
- Collaborate with engineering and AI teams through GitHub, code reviews, and weekly sprints, delivering clean, well-documented code and proposing new feature and AI enhancements.

Auburn University of Montgomery Learning Center

August 2025 – Present

Mathematics & Computer Science Tutor

Montgomery, AL

- Deliver one-on-one and group tutoring sessions for **20–25** students weekly, contributing to an average **10–20%** improvement in exam and assignment performance among returning students.
- Create and maintain a respectful, inclusive learning environment, resulting in consistent **90%+** student return rates for scheduled tutoring sessions.
- Collaborate with staff to track student progress and communicate academic concerns, enabling more targeted interventions and contributing to higher overall student success rates.

Technical Projects

Wapi 🌐 — Restaurant Discovery & Community Platform | React, Supabase (PostgreSQL), Capacitor, Node.js

- Founded and solo-built a web + mobile restaurant-discovery platform for Kinshasa (V1 launched May 2026), ingesting **1,888** restaurants from scratch via a Google Places + OpenStreetMap pipeline with trigram + Haversine deduplication.
- Built a custom reverse-geocoder (**24** GPS commune centroids, nearest-centroid) that re-assigned **77%** of records with unusable location data, populating **22 of 24** communes at **\$0** cost vs. a paid re-scan.
- Shipped real-time social feeds (Supabase Realtime) and a single React codebase deployed to web (Vercel CI/CD) and mobile (Capacitor, iOS + Android).

AI Website Builder 🌐 — Full-Stack SaaS | React, TypeScript, Node.js, Express, PostgreSQL, Prisma, Stripe

- Built an AI tool that turns a plain-English prompt into a complete website, streaming it live in the browser so users watch their page build in real time — replacing a sluggish **4s** wait with instant feedback.
- Built a signature-verified, idempotent Stripe webhook flow that grants credits only after confirmed payment, eliminating failed-payment credit loss.
- Built the full backend — secure user accounts and **15+** REST endpoints over a **7-table** PostgreSQL database — powering project versioning, one-click rollback, and a chat-based AI editing loop.

PunchIQ 🌐 — End-to-End ML Fight Prediction System | Python, FastAPI, scikit-learn, SHAP, React, TypeScript

- Built and deployed an end-to-end ML system predicting boxing outcomes at **80.9%** test accuracy, beating a Logistic Regression baseline by **+2.9pp** and a naive baseline by **+30.9pp**.
- Engineered a **7-feature** pipeline with symmetric data augmentation that doubled samples to **725K**, and prevented data leakage by splitting before encoding and fitting the scaler on the training set only.
- Benchmarked **3** models (Logistic Regression, HistGradientBoosting, Random Forest) and deployed the winner as a live FastAPI web app — slimmed to **6.5 MB** to run on a free **512 MB** server — that explains every prediction in plain terms with SHAP.